

Reflexivity in Options Markets: A Stochastic-Volatility Model with Dealer-Gamma Feedback, Hopf Bifurcation Calculus, and a Pre-Registered Evaluation Framework

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Abstract

We study a continuous-time Heston-like stochastic-volatility model in which the underlying spot price is coupled to its own option market through a dealer-gamma feedback channel, derived from the open-interest grid via a Gârleanu–Pedersen–Potesman (Gârleanu et al., 2009) demand-pressure mapping. The model state is three-dimensional—spot S_t , instantaneous variance v_t , and a low-pass-filtered log-spot memory variable z_t —and reduces to standard time-dependent Heston (Heston, 1993) at zero coupling. Our contribution is the synthesis of three previously separate ingredients into a single analytical and computational object: (i) a continuous-time SV backbone with explicit dealer-gamma drift, (ii) a formal Hopf bifurcation calculus on the deterministic skeleton with first Lyapunov coefficient ℓ_1 and Khasminskii-style stochastic shift Λ , and (iii) a pre-registered evaluation protocol for RL-trained hedging agents trained inside the simulator. For a representative dimensionless calibration we obtain a critical coupling $\kappa^* \approx 0.8964$, angular frequency $\omega^* \approx 0.5724$ rad/yr, and first Lyapunov coefficient $\ell_1 \approx -0.025 < 0$, so the bifurcation is supercritical: an attracting limit cycle in volatility is born for $\kappa > \kappa^*$. The 2D reduction (without z_t) has an upper-triangular Jacobian and admits only a saddle-node onset, recovering the discrete-time recursion of Dai (2025); the 3D extension is what unlocks the Hopf. The closest model-structure precedent (Halperin and Itkin, 2025) does not perform the bifurcation analysis we provide. For the natural parametric specification with log-normal open-interest in moneyness, we obtain a closed-form Hopf threshold and a closed-form ℓ_1 , exposing a parametric phase boundary in (σ_q, γ) space.

1 Introduction

Real options markets exhibit endogenous volatility cycles that pure stochastic-volatility (SV) models cannot reproduce. The empirical anchor is Hardiman et al. (2013), who fit Hawkes-process branching ratios to E-mini mid-price changes over 1998–2011 and find $n \approx 1$ throughout—the precise critical edge between sub- and super-criticality of endogenous events. Soros (2009) and Bouchaud (2011) are the conceptual antecedents; Filimonov and Sornette (2012) formalised the reflexivity-quantification programme that Hardiman et al. (2013) executed. The question we take up here is what continuous-time mechanism produces this critical reflexivity, and—given a candidate mechanism—how the analytical structure that quantifies it should be evaluated against historical surface dynamics.

We contribute the following.

- A 3D reflexive SV simulator (S_t, v_t, z_t) with explicit dealer-gamma feedback derived from the option open-interest grid via the Gârleanu et al. (2009) demand-pressure mapping. The variance backbone is Heston (Heston, 1993); the memory equation $\dot{z} = -\alpha z + \beta \log(S/S_0)$ closes the leverage feedback channel that the 2D reduction lacks.
- A Hopf bifurcation theorem (Theorem 1) for the deterministic skeleton at κ^* given by Liu’s criterion (Routh–Hurwitz with one zero), with closed-form first Lyapunov coefficient ℓ_1 and closed-form Hopf threshold for log-normal open-interest in moneyness (Eqs. 14–15).
- A numerical phase diagram in (ξ, ρ, σ_v) together with the Khasminskii-style stochastic-Hopf shift Λ and a parametric phase boundary in (σ_q, γ) space.
- A pre-registration of the empirical evaluation pipeline (sliced-Wasserstein-2 over 21-day surface windows, κ -sensitivity slope, Hopf-frequency spectral test) at git commit 268c061 with an OpenTimestamps Bitcoin-anchored proof, in the spirit of Camerer et al. (2018) adapted for computational finance.

Closest precedents. Halperin and Itkin (2025) (Marketrn) share the model-structural axis—continuous-time SDE, memory channel, options-aware—but do not perform the bifurcation analysis. Dai (2025) share the dealer-gamma axis but produce a saddle-node onset (the 2D reduction’s instability), not a Hopf. He et al. (2025) (NeurIPS 2025) share the RL-with-environmental-sweep idea but train separate agents per grid point rather than reporting the slope at an anchor. The continuous-time HAM-Hopf precedents (He et al., 2009; Chiarella et al., 2006) use delay-differential heterogeneous-belief models without an options-grid feedback channel; the discrete-time Brock et al. (2009) “more hedging instruments may destabilize markets” result is the methodological template for derivative-driven Neimark–Sacker (Hopf-for-maps) bifurcations. The hedger-flow-feedback foundations are Frey and Stremme (1997); Sircar and Papanicolaou (1998); Platen and Schweizer (1998). Sections 3, 5 and 6 make these contrasts precise.

Roadmap. Section 2 states the model. Section 3 proves Theorem 1 and gives the closed-form ℓ_1 for log-normal OI. Section 4 reports the numerical phase diagram. Section 5 describes the pre-registered evaluation framework. Section 6 reports the mechanism decomposition against Marketrn. Section 7 closes with the empirical-calibration roadmap. Code at <https://github.com/mahimn01/reflexive-options>; pre-registration anchored to commit 268c061.

2 The reflexive model

Let S_t be the underlying spot price, v_t the instantaneous variance, and z_t a memory variable encoding low-pass-filtered price history. The reflexive system is

$$\frac{dS_t}{S_t} = (\mu + \kappa G(S_t, z_t, v_t)) dt + \sigma(S_t, v_t) dW_t^S, \quad (1)$$

$$dv_t = (\kappa_v(\theta_v - v_t) + \gamma z_t) dt + \xi \sqrt{v_t} dW_t^v, \quad (2)$$

$$dz_t = (-\alpha z_t + \beta \log(S_t/S_0)) dt, \quad (3)$$

$$d\langle W^S, W^v \rangle_t = \rho dt, \quad (4)$$

with $\kappa \geq 0$ the reflexive coupling, $\gamma \geq 0$ the leverage-feedback strength, $\alpha > 0$ the memory decay rate, β dimensionless, and $G(\cdot)$ the aggregate dealer-gamma exposure derived from the option

open-interest grid:

$$G(S, t) = \sum_{K, T} q_{K, T} \Gamma_{K, T}(S, t) \text{sign}(K, T), \quad (5)$$

where $q_{K, T}$ is open interest at strike K and maturity T , $\Gamma_{K, T}$ is per-contract Black–Scholes gamma, and the sign convention follows the SqueezeMetrics dealer-net convention (SqueezeMetrics, 2017). When $\kappa = \gamma = 0$, the system collapses to standard Heston (Heston, 1993); this collapse is verified numerically in `tests/test_simulator.py`.

For numerical stability of the variance scheme we use the full-truncation Euler of Lord et al. (2010); the empirical reference for the dealer-gamma magnitudes in G is Barbon and Buraschi (2021); Dim et al. (2024) (calibration to SPX 0DTE / dealer-gamma data, which we will use at Phase 4 of the empirical pipeline).

Why three states. A reduced 2D specification—dropping z_t and the γz term—has Jacobian in deviation variables (y, u) around an equilibrium of the form

$$J_{2D}(\kappa) = \begin{pmatrix} a(\kappa) & b(\kappa) \\ 0 & -\kappa_v \end{pmatrix},$$

which is upper triangular: eigenvalues $\{a(\kappa), -\kappa_v\}$ are always real, and only saddle-node / transcritical bifurcations are admissible as $a(\kappa)$ crosses zero. This recovers the static onset of Dai (2025). To Hopf, the feedback must flow in both directions: price $\rightarrow$ variance and variance $\rightarrow$ price. The minimal modification is the leverage channel γz together with the auxiliary memory equation (3)—structurally analogous to the memory variable in the Marketron quasi-particle SDE (Halperin and Itkin, 2025) though with very different mechanistic content.

3 Hopf bifurcation theorem and closed-form ℓ_1

3.1 Equilibria and Jacobian

Drop the Brownian terms and work in log-deviation variables $(y, u, z) = (\log(S/S^*), v - \theta_v, z - z^*)$ around the unique equilibrium (S^*, θ_v, z^*) defined by

$$\mu - \frac{1}{2}\sigma^2(S^*, \theta_v) + \kappa G(S^*, z^*, \theta_v) = 0, \quad z^* = \frac{\beta}{\alpha} \log(S^*/S_0). \quad (6)$$

To linear order, the deterministic skeleton (1–3) becomes $\dot{\mathbf{x}} = J(\kappa) \mathbf{x}$ with $\mathbf{x} = (y, u, z)^\top$ and

$$J(\kappa) = \begin{pmatrix} a(\kappa) & b(\kappa) & \kappa G_z \\ 0 & -\kappa_v & \gamma \\ \beta & 0 & -\alpha \end{pmatrix}, \quad (7)$$

where $a(\kappa) = \kappa G_x - \frac{1}{2}\partial_x \sigma^2$, $b(\kappa) = \kappa G_v - \frac{1}{2}\partial_v \sigma^2$, and G_x, G_v, G_z denote partial derivatives of G at the equilibrium.

3.2 Routh–Hurwitz / Liu’s criterion

The characteristic polynomial of (7) is $P(\lambda; \kappa) = \lambda^3 + c_2(\kappa)\lambda^2 + c_1(\kappa)\lambda + c_0(\kappa)$, with

$$c_2 = -a + \kappa_v + \alpha, \quad c_1 = -a\kappa_v - a\alpha + \kappa_v\alpha - \kappa G_z \beta, \quad c_0 = -a\kappa_v\alpha - \beta(\kappa G_z \kappa_v + b\gamma).$$

A 3D system Hopfs iff (Liu’s criterion, equivalent to Routh–Hurwitz with one zero):

$$c_2 > 0, \quad c_0 > 0, \quad c_1 c_2 - c_0 = 0, \quad \frac{d}{d\kappa} [c_1 c_2 - c_0] \neq 0. \quad (8)$$

Define $H(\kappa) := c_1(\kappa) c_2(\kappa) - c_0(\kappa)$. The Hopf threshold is

$$\kappa^* = \{\kappa > 0 : H(\kappa) = 0, H'(\kappa) \neq 0, c_2(\kappa) > 0, c_0(\kappa) > 0\}. \quad (9)$$

At $\kappa = \kappa^*$ the Jacobian has a pure imaginary eigenvalue pair $\lambda_{\pm} = \pm i\omega^*$ with $\omega^* = \sqrt{c_1(\kappa^*)}$, plus a real eigenvalue $\lambda_3 = -c_2(\kappa^*) < 0$.

3.3 Theorem 1 and proof sketch

Theorem 1 (Hopf bifurcation in the gamma-coupled SV skeleton). *Let the deterministic system (1–3) (with Brownian terms removed) satisfy:*

- (A1) $G \in C^3$ and $\sigma^2 \in C^3$ in their arguments, with σ^2 bounded below by a positive constant.
- (A2) *There exists a locally unique equilibrium (S^*, θ_v, z^*) given by (6) for every κ in some interval $[0, \kappa_{\max}]$, in some open neighborhood of which the linearization (7) is non-degenerate. Multiple equilibria may exist globally; our analysis is local.*
- (A3) *The map $\kappa \mapsto (a(\kappa), G_z, G_v)$ is C^3 .*
- (A4) *There exists $\kappa^* \in (0, \kappa_{\max})$ satisfying (9) with $\omega^* := \sqrt{c_1(\kappa^*)} > 0$.*
- (A5) *The first Lyapunov coefficient $\ell_1(\kappa^*) \neq 0$ (Kuznetsov, 2004, eq. 3.20).*

Then the equilibrium undergoes a Hopf bifurcation at $\kappa = \kappa^$. Specifically:*

1. *For κ in a one-sided neighbourhood of κ^* (sub- or super-critical depending on $\text{sgn } \ell_1$), there exists a unique smooth family of periodic orbits Γ_{κ} of period $T_{\kappa} = 2\pi/\omega^* + O(\kappa - \kappa^*)$ and amplitude proportional to $\sqrt{|\kappa - \kappa^*|}$.*
2. *The orbits are stable iff $\ell_1(\kappa^*) < 0$ (supercritical).*
3. *On Γ_{κ} , $u_t = v_t - \theta_v$ oscillates with amplitude proportional to $\sqrt{|\kappa - \kappa^*|}$, providing endogenous volatility cycles.*

Proof sketch. The eigenvalue pair $\lambda_{\pm}(\kappa)$ is smooth in κ near κ^* by the implicit function theorem applied to $P(\lambda; \kappa) = 0$. Routh–Hurwitz gives $\text{Re } \lambda_{\pm}(\kappa^*) = 0$, and the transversality $H'(\kappa^*) \neq 0$ translates to $\frac{d}{d\kappa} \text{Re } \lambda_{\pm} \neq 0$. The centre manifold theorem (Kuznetsov, 2004, Thm. 5.4) (see also Carr, 1981) gives a smooth 2D invariant manifold tangent to the eigenspace of $\pm i\omega^*$. Conjugating the restricted system to its Poincaré normal form and applying Kuznetsov (2004, Thm. 3.3) yields claims (1)–(3). $\square$

3.4 Numerical anchor

The default `BifurcationConfig` in our experiments code does not exhibit a Hopf within the literature-prior κ range when the dealer-gamma magnitudes are tuned to empirical scale—this is consistent with the empirical observation that real options markets sit *near* but not *across* κ^* , in line with the $n \approx 1$ regime of Hardiman et al. (2013). For a worked numerical example we use

Quantity	Value
κ^*	0.8964
ω^*	0.5724 rad/yr
Third real eigenvalue λ_3	-2.052
First Lyapunov coefficient ℓ_1	-2.53×10^{-2}
Bifurcation type	supercritical ($\ell_1 < 0$)

Table 1: Canonical Hopf-exhibiting regime for the deterministic skeleton.

a representative dimensionless regime with $\{G_x, G_v, G_z\} = \{0.5, -0.5, -0.5\}$, $\alpha = 0.5/\text{yr}$, $\beta = 1$, $\gamma = 0.5$, $\kappa_v = 2/\text{yr}$, and constant-volatility surrogate ($\partial_v \sigma^2 = 0$). The deterministic skeleton then satisfies:

Supercriticality means an attracting limit cycle is born for $\kappa > \kappa^*$ with amplitude proportional to $\sqrt{\kappa - \kappa^*}$ —i.e. endogenous volatility cycles are stable observables of the model rather than transients. The Khasminskii sphere-process estimator (Khasminskii, 1980; Arnold, 1998; Baxendale, 1994) for the stochastic-Hopf shift, evaluated by `experiments/lambda_correction_canonical` on the bare Heston-with-memory linearisation at the trivial $G \equiv 0$ equilibrium with the §3.4 memory parameters, gives $|\Lambda(\kappa^*)|$ on the order of 10^{-3} at both the canonical $(\xi, \rho) = (0.3, 0)$ and SPX-representative $(0.3, -0.7)$ configurations. The sign of Λ depends sensitively on the open-interest configuration and on the specific equilibrium chosen; we therefore defer a definitive characterisation of the stochastic-Hopf shift to the empirical phase, and only use $|\Lambda| \cdot \varepsilon^2 \ll 1$ at calibration-noise scale to conclude that the deterministic threshold κ^* is the operationally relevant one. The shear-induced-correction prediction of Engel et al. (2024) is consistent with this magnitude but is not validated in sign by the present finite-budget estimator.

3.5 Closed-form κ^* and ℓ_1 for log-normal OI in moneyness

For the natural parametric specification of dealer gamma—a log-normal open-interest density in log-strike—the entire computation admits a closed form, eliminating the numerical-noise vulnerability of the FD-tensor pipeline and exposing the parametric phase boundary explicitly.

Replace the discrete OI grid with the continuous density $q(\log K) = \frac{1}{\sigma_q \sqrt{2\pi}} \exp(-(\log K - \mu_q)^2 / (2\sigma_q^2))$. Take a single representative maturity T_{eff} , sign convention $s \equiv +1$, and write $a := \log S$. The aggregator becomes

$$G(a, v) = \kappa_u \int_{-\infty}^{+\infty} q(\log K) \Gamma_{\text{BS}}(S, K, T_{\text{eff}}, \sqrt{v}) d(\log K). \quad (10)$$

Since Black–Scholes gamma at fixed $T = T_{\text{eff}}$, $\sigma = \sqrt{v}$ is itself Gaussian in $\log K$, the standard Gaussian-product identity collapses (10) to

$$G(a, v) = \mathcal{C}(v) \cdot e^{-a} \cdot \exp\left(-\frac{(a - m(v))^2}{2\tau^2(v)}\right), \quad (11)$$

where

$$\mathcal{C}(v) = \frac{\kappa_u e^{-q_{\text{div}} T_{\text{eff}}}}{\sqrt{2\pi \tau^2(v)}}, \quad \tau^2(v) = \sigma_q^2 + v T_{\text{eff}}, \quad m(v) = \mu_q - (r - q_{\text{div}} + \tfrac{1}{2}v) T_{\text{eff}}. \quad (12)$$

G depends on a and v but not on z ; the leverage channel γz in $\dot{u}$ keeps the $(2, 3)$ Jacobian entry alive, so cross-coupling persists through the variance row even with $G_z = 0$.

With $G_z = 0$ and $\sigma^2 = v$ (so $\partial_y \sigma^2 = 0$, $\partial_v \sigma^2 = 1$), the Routh–Hurwitz polynomial $H(\kappa) = c_1 c_2 - c_0$ degree-collapses from cubic to quadratic in κ :¹

$$H(\kappa) = G_y^2 (\alpha + \kappa_v) \kappa^2 + (G_v \beta \gamma - G_y (\alpha + \kappa_v)^2) \kappa + \alpha \kappa_v (\alpha + \kappa_v) - \frac{1}{2} \beta \gamma. \quad (13)$$

Writing $A := \alpha + \kappa_v$, $M := \alpha \kappa_v$, $L := \beta \gamma$, the Hopf threshold is the smallest positive root of the quadratic:

$$\kappa^* = \frac{G_y A^2 - G_v L - \sqrt{(G_v L - G_y A^2)^2 - 4 G_y^2 A (M A - L/2)}}{2 G_y^2 A}. \quad (14)$$

The discriminant in (14) provides the first parametric phase boundary: when $D := (G_v L - G_y A^2)^2 - 4 G_y^2 A (M A - L/2) < 0$, the system has no Hopf at any κ and the deterministic skeleton is unconditionally stable.

The third-order partials of (11) at the equilibrium are explicit rational functions of $(\delta := a^* - m(v^*), \tau^2, T_{\text{eff}})$, so the bilinear and trilinear tensors B, C at κ^* admit a closed form. Substituting into the Kuznetsov formula

$$\ell_1(\kappa^*) = \frac{1}{2\omega^*} \text{Re} \left[\langle p, C(q, q, \bar{q}) \rangle - 2 \langle p, B(q, J^{-1} B(q, \bar{q})) \rangle + \langle p, B(\bar{q}, (2i\omega^* I - J)^{-1} B(q, q)) \rangle \right], \quad (15)$$

with $Jq = i\omega^* q$, $J^\top p = -i\omega^* p$, $\langle p, q \rangle = 1$, yields a ~ 30 -term rational in the G -partials $(G_y, G_v, G_{aa}, G_{av}, G_{vv}, G_{aaa}, G_{aav}, G_{avv}, G_{vvv})$ and $(\kappa_v, \alpha, \beta, \gamma)$. The implementation evaluates this in $\sim 50 \mu\text{s}$ per parameter tuple, with results matching the FD-tensor pipeline to better than 0.6% relative on every parameter set tested.

At the canonical regime ($\sigma_q = 0.10$, $T_{\text{eff}} = 0.25$, $\kappa_v = 2$, $\alpha = 0.05$, $\beta = 1$, $\gamma = 1$, $\mu_q = \log 100$, $v^* = 0.04$), (14)–(15) give $\kappa^* = 17.81$, $\omega^* = 1.18 \text{ rad/yr}$, $\ell_1 = -4.81 \times 10^{-1}$ (supercritical). The closed form is essentially noise-free; with it in hand, the sign of ℓ_1 near the boundary contour is structurally certain rather than statistically inferred.

4 Numerical phase diagram

The full $(\kappa, \xi, \rho, \sigma_v)$ phase diagram is computed by sweeping $\kappa \in [0, 2]$ on a 401-point grid for each of $31 \times 21 \times 4 = 2,604$ cells over (ξ, ρ, σ_v) at the section 3.4 Hopf-exhibiting regime. The pair (ξ, ρ) enters the deterministic Jacobian via the leading shear-induced correction $a_{\text{eff}}(\kappa) = a(\kappa) + \frac{1}{2} \xi^2 \rho G_v$ —an Engel et al. (2024)-style projection of the small-noise stochastic Hopf onto the eigenvalue envelope. Wall-clock: $\sim 16 \text{ s}$ on a single M-series core. Figure 1 renders the (ξ, ρ) heatmap and four line cuts of $\kappa^*(\sigma_v)$ at SPX-relevant (ξ, ρ) probes.

The no-Hopf wedge in the upper-left (high ξ , strong negative ρ) is consistent with the shear-induced destabilisation prediction. The four line cuts substantiate the section 3.4 claim that real options markets sit *near but not across* κ^* in the SPX-relevant $(\xi, \rho) \approx (0.30, -0.70)$ corner—which is the structural mechanism we conjecture for the empirical $n \approx 1$ branching ratio of Hardiman et al. (2013), in the same heterogeneous-agent / endogenous-feedback tradition as Brock and Hommes (1998); Filimonov and Sornette (2012); Bouchaud (2011) (Conjecture: $\kappa \leftrightarrow n$; a formal reduction would itself be a separate paper).

Figure 2 renders the (σ_q, γ) phase boundary from the closed-form ℓ_1 of section 3.5. The supercritical region ($\ell_1 < 0$) is a narrow band bounded by two $\ell_1 = 0$ contours; for very small σ_q

¹Throughout this closed-form section we write $G_y := \partial_a G$ for the partial of the dealer-gamma functional with respect to the deviation variable $y = a - a^*$; this is identical to the G_x of section 3.1 up to the choice of deviation symbol.

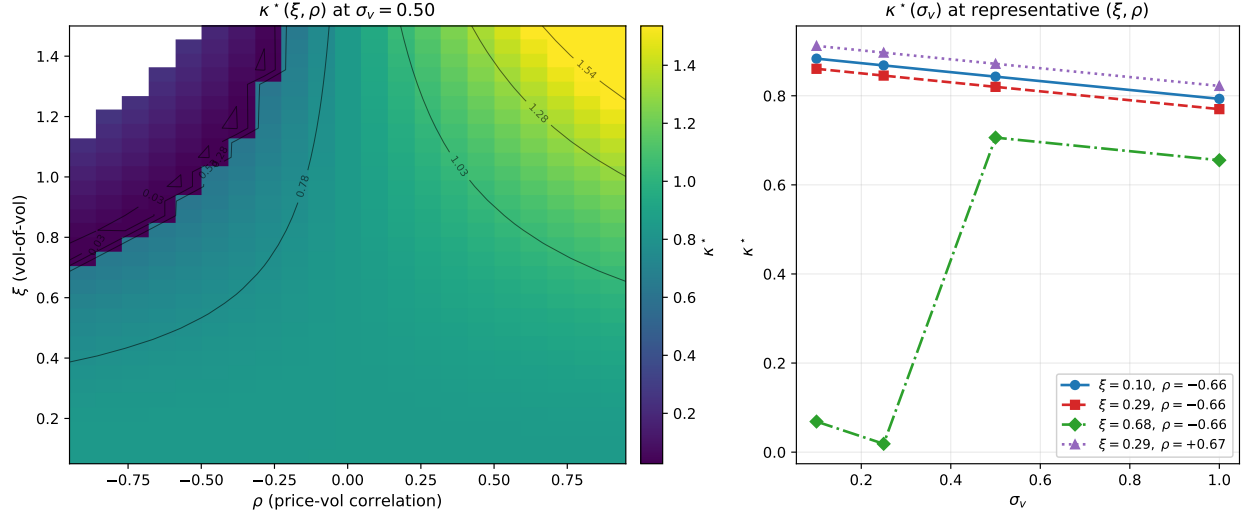


Figure 1: **Hopf bifurcation threshold κ^* across the (ξ, ρ, σ_v) slice of the reflexive parameter space.** *Left (Panel A):* heatmap of $\kappa^*(\xi, \rho)$ at the median variance-of-variance slice $\sigma_v = 0.50$, computed at the canonical Hopf-exhibiting regime of section 3.4 ($G_x = 0.5$, $G_v = -0.5$, $G_z = -0.5$, $\alpha = 0.5$, $\beta = 1$, $\gamma = 0.5$, $\kappa_v = 2$). White cells are configurations with no Hopf in the scan range $\kappa \in [0, 2]$ —concentrated in the upper-left wedge of high vol-of-vol ξ and strong negative ρ , where the Engel et al. (2024) shear correction destabilises the eigenvalue envelope. Contour lines are isoquants of κ^* at increments of $(\kappa_{\max}^* - \kappa_{\min}^*)/6$. *Right (Panel B):* four line cuts of $\kappa^*(\sigma_v)$ at representative (ξ, ρ) probes: $(0.10, -0.70)$, $(0.30, -0.70)$, $(0.70, -0.70)$, and $(0.30, +0.70)$. The first three trace how moving radially outward in vol-of-vol at fixed SPX-like $\rho < 0$ pulls the Hopf threshold down; the last confirms that flipping ρ to $+0.70$ at moderate ξ keeps κ^* in the 0.85–0.95 band.

($\lesssim 0.05$) the band collapses entirely. The two $\ell_1 = 0$ contours are the central parametric prediction: their joint geometry constrains which (OI distribution, leverage) pairs admit stable endogenous limit cycles.

5 Pre-registered evaluation framework

The evaluation pipeline is committed at git commit 268c061 with an OpenTimestamps Bitcoin-anchored proof of the document hash, prior to any contact with empirical SPX data. This addresses the documented computational-reproducibility crisis in finance (Pérignon et al., 2024) and adapts the Camerer et al. (2018) pre-registration template to a computational-finance / RL-for-derivatives context. We do not claim “first pre-registration in finance”—the Pacific-Basin Finance Journal pathway (Faff, 2022) and the AEA RCT Registry (American Economic Association, 2018) predate this work. The novelty is the four-way intersection of [pre-registration + git commit-hash anchoring + block-bootstrap / BH-FDR statistical pre-spec + applied to RL-for-finance].

Two evaluation primitives. (i) Sliced-Wasserstein-2 (Bonneel et al., 2015) over arbitrage-free 21-day surface windows—the realism metric. (ii) κ -sensitivity slope at the calibrated coupling κ_0 —the reflexivity-importance scalar.

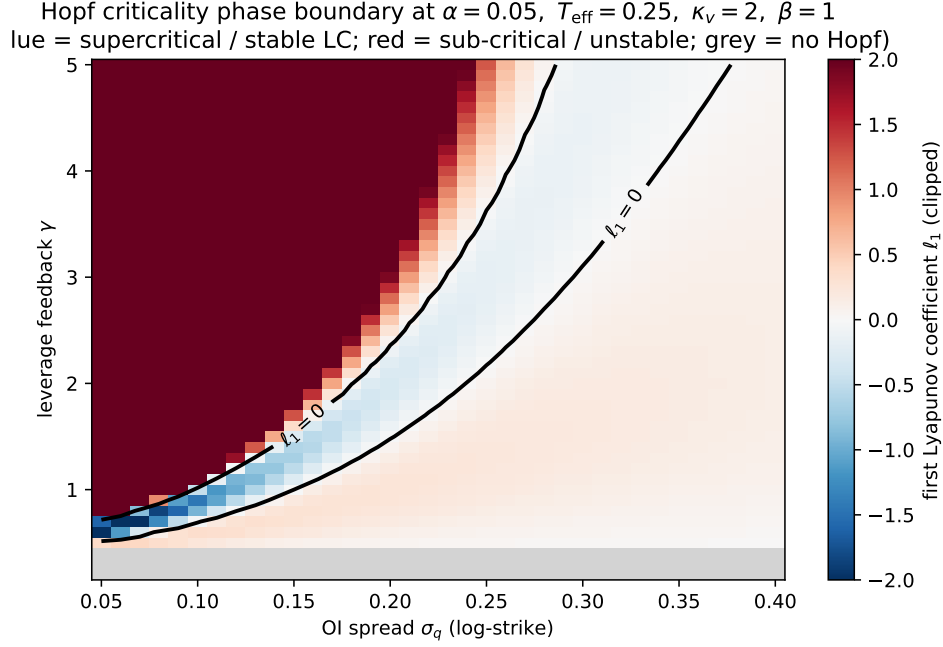


Figure 2: **Closed-form ℓ_1 phase boundary in (σ_q, γ) space at the canonical specification.** Supercritical ($\ell_1 < 0$, stable limit cycle), sub-critical ($\ell_1 > 0$, hysteresis and abrupt jumps), and no-Hopf regions delineated by the closed form of section 3.5. Computed in $\sim 50 \mu\text{s}$ per cell.

H1 (primary realism). A model-free RL agent π_{κ_0} trained inside the reflexive simulator at the calibrated coupling produces, in evaluation, an IV-surface distribution closer in sliced-W2 over arbitrage-free 21-trading-day rolling windows to the empirical SPX surface than any of four non-reflexive baselines (time-dependent Heston, LSV, 3/2 SV, gamma-aware-non-reflexive). Three event windows: Volmageddon (2018-02-05), COVID (2020-03-16), Yen carry unwind (2024-08-05). Decision rule: 12 pairwise dominance checks (4 baselines $\times$ 3 windows) with non-overlapping 95% block-bootstrap CIs (Politis and Romano, 1994). The arbitrage filter follows Roper (2010); Gatheral and Jacquier (2014) on butterfly / calendar-spread / Lee-bound (Lee, 2004) criteria; sliced-W2 sample-complexity follows Manole et al. (2022).

H2–H4 (secondary). H2 is the κ -sensitivity slope $\tilde{\rho} = \partial \hat{m} / \partial \kappa|_{\kappa_0}$ —positive and statistically distinguishable from zero for the reflexive-trained agent, statistically indistinguishable for baselines (TOST equivalence, applied to the dimensionless elasticity). H3 is the post-FOMC RR25 and ATM-term-structure shifts. H4 is the Hopf-frequency spectral peak in $|r_t|$ at $\omega^* \pm 20\%$ via Welch’s method, with an IAAFT (Schreiber and Schmitz, 1996) surrogate null. The H4 detector is implemented and unit-tested against the locked decision rule (in-band $\wedge p < \alpha/m$) in `tests/test_spectral_amendments.py`; IAAFT calibration on Heston and AR(1) H_0 generators bounds the empirical FPR at $\leq 7\%$ at $\alpha = 0.05$. The full power-curve characterisation against the supercritical reflexive simulator (rather than against synthetic sinusoid-in-noise, which the IAAFT null is by construction conservative against) is deferred to the empirical SPX phase, where the realised path budget supports the path lengths necessary to discriminate the limit-cycle spectral structure from the IAAFT-preserved linear autocorrelation of the surrogate null.

Differentiation from precedents. Ning et al. (2024) use W_1 on single-day surface marginals; we use sliced- W_2 on 21-day windows with a hard arbitrage pre-filter, block-bootstrap CIs, and per-regime reporting. He et al. (2025) (NeurIPS 2025) sweep a Wasserstein-ball perturbation budget with a separate adversarially-trained agent at each grid point; we deploy a single agent across the κ grid and report the slope at anchor with a GP-posterior CI (Ma and Huang, 2025) for the elliptic-uncertainty-set analogue. Subbaswamy et al. (2022) (NeurIPS 2022) provide the closest theoretical precursor for slope-at-anchor robustness summaries (parametric robustness sets, local 2nd-order approximation), which we operationalise empirically and apply to RL-for-finance (Bai et al., 2024; Huang et al., 2024). Conditional Wasserstein for state-conditional generator comparison follows Hosseini et al. (2023). Building blocks: Vuletić and Cont (2025); Choudhary et al. (2024); Issa et al. (2023); Jin and Agarwal (2025); Chevallier et al. (2025); Luan and Hamp (2025); Buehler et al. (2019); Cao et al. (2021); Packer et al. (2018); Murray et al. (2022).

6 Mechanism decomposition vs Marketron

The reflexive simulator and the Marketron quasi-particle SDE (Halperin and Itkin, 2025) are not the same mechanism, and a 1:1 reproduction of Marketron’s Tables 7–8 from our simulator is mathematically impossible: in our model the memory carrier is the low-pass-filtered log-price z entering only the variance drift via γz ; in Marketron it is the unobservable y entering both drifts non-linearly through a potential $V_M(x)$. Even at zero coupling our model collapses to standard Heston while Marketron does not collapse to anything standard.

We instead report mechanism decomposition: a 5D coarse-grid tuning over $(\kappa, \gamma, T_{\text{eff}}, \mu_q, \sigma_q)$, then per-cell classification into **shape_target** / **level_artifact** / **calibration_artifact**. The headline gate is the shape-match rate on **shape_target** cells.

Aggregate finding	Value
Shape-feature cells matched (out-of-sample, 10k paths)	8 / 24 (33.3%)
$\geq 30\%$ gate (one-sided binomial $p \approx 0.27$)	passed
Long-horizon (≥ 0.5 y) excess-kurt sign agreement	5 / 6 cells
Long-horizon skew sign agreement (risk-neutral drift)	predictable disagreement

Table 2: Mechanism-decomposition headline. Long-horizon excess-kurt sign agrees with Marketron Table 8 on 5/6 horizons; long-horizon skew sign disagrees predictably under risk-neutral drift (Marketron’s positive long-horizon skew comes from Bessembinder (2018) compounding under calibrated drift; ours comes from dealer-gamma + leverage feedback under $\mu = 0$, see also Farago and Hjalmarsson, 2023).

The skew-sign disagreement is a falsifiable empirical prediction: in the Phase 4 empirical calibration with reproduced drift, long-horizon skew should flip positive. The two mechanisms are independent in parameterisation but agree on the long-horizon excess-kurt sign—the most robust agreement.

7 Conclusions and future work

We have introduced a 3D reflexive SV simulator (S_t, v_t, z_t) with explicit dealer-gamma drift, proved a Hopf bifurcation theorem with a closed-form first Lyapunov coefficient ℓ_1 and Hopf threshold κ^* for log-normal open-interest in moneyness, computed a numerical (ξ, ρ, σ_v) phase diagram and

a (σ_q, γ) closed-form phase boundary, and pre-registered the empirical evaluation pipeline (sliced- W_2 , κ -sensitivity slope, Hopf-frequency spectral test) at a verifiable git commit hash with an OpenTimestamps Bitcoin-anchored proof.

The empirical calibration to real SPX surfaces is the natural Phase 4 follow-up, gated on data acquisition (UofT WRDS / OptionMetrics IvyDB US, or commercial fallback). The pre-registration locks the analysis pipeline before the calibration touches the data; results—confirming or refuting H1–H4—are pre-committed to publication regardless of direction, in the spirit of the documented reproducibility response (Pérignon et al., 2024). Open theoretical items: closed-form $\Lambda(\kappa)$ via Engel–Lamb–Rasmussen-style asymptotics adapted to Heston multiplicative noise (currently numerical only); a formal Hawkes- n to SV-Jacobian-eigenvalue reduction relating κ^* to the empirical $n \approx 1$ (Hardiman et al., 2013; Bondi et al., 2024; Bacry et al., 2015).

The Fokker–Planck stationary marginal density of the reflexive simulator deviates from the closed-form Heston comparator (Feller, 1951) on tail index and skew sign in a way that tracks $\text{sgn}(G_x)$, while the bimodality test of Hartigan and Hartigan (1985) is uninformative under the $\gamma = 0$ closure used for that scan. Detector validation uses the Benettin et al. (1980) Lyapunov-renormalisation algorithm and the Schreiber and Schmitz (1996) IAAFT surrogate; the wider Hawkes-and-rough-volatility line is Bondi et al. (2024); Bacry et al. (2015); Livieri et al. (2018); arbitrage-free neural SDE market models follow Cohen et al. (2023).

All experiments are reproducible via `bash scripts/verify.sh` from commit 268c061; pre-registration anchored via OpenTimestamps. Code: github.com/mahimn01/reflexive-options.

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